

Module 1

DC Circuit Analysis and Theorems

Dr. Elizabeth Rufus

Professor, School of Electronics Engineering

Maximum Power Transfer Theorem

In practical situations, a circuit is designed to provide power to a load.

There are applications in areas such as communications where it is desirable to maximize the power delivered to a load.

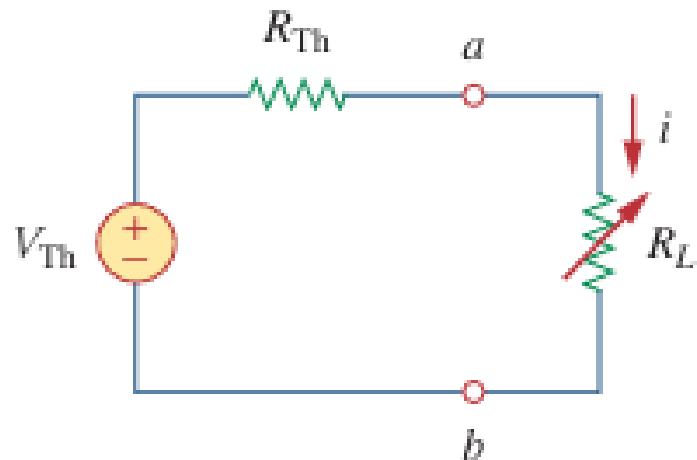
We have to address the problem of delivering the maximum power to a load when given a system with known internal losses.

The Thevenin's equivalent is useful in finding the maximum power a linear circuit can deliver to a load.

We assume that we can adjust the load resistance.

If the entire circuit is replaced by its Thevenin's equivalent except for the load, the power delivered to the load is,

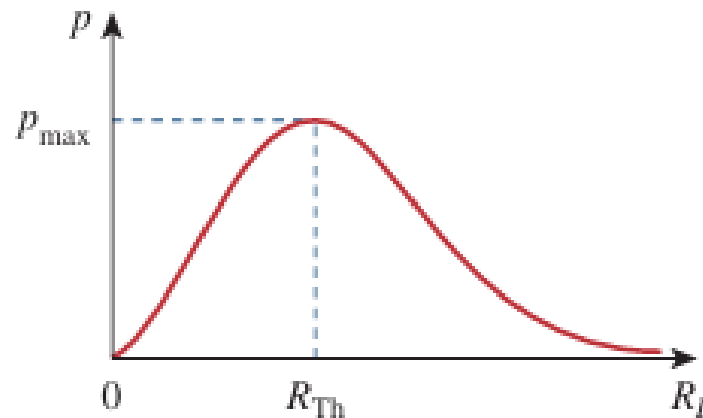
$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$



Maximum Power Transfer Theorem

V_{th} and R_{th} are fixed and by varying the load resistor R_L the power delivered to the load can be varied.

We notice that the power is small for small or large values of R_L but maximum for some value of R_L between 0 and ∞ .



The **maximum power** occurs when $R_L = R_{th}$. This is known as the **maximum power theorem**.

Maximum Power Transfer Theorem

To prove Maximum power transfer theorem differentiate with respect to R_L and set the result equal to zero:

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L \quad \text{----- 1}$$

$$\frac{dp}{dR_L} = V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right]$$

$$0 = (R_{Th} - R_L)$$

$$R_L = R_{Th}$$

This shows that the maximum power transfer takes place when the load resistance equals the Thevenin resistance ($R_L = R_{th}$)

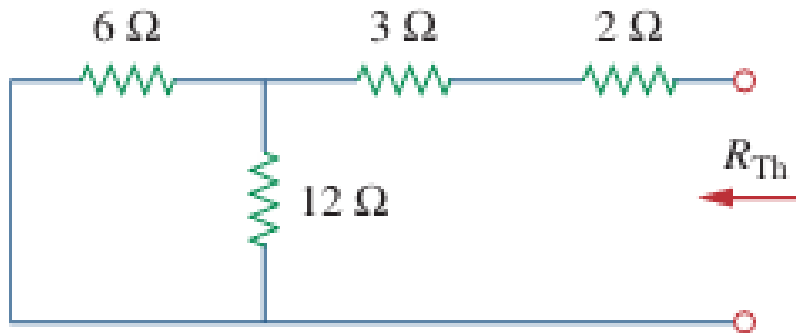
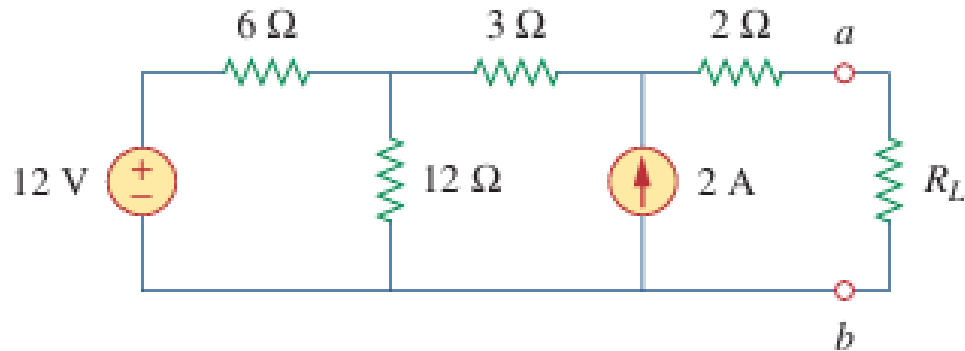
Hence, the maximum power transferred is:

$$p_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

This equation applies only when $R_L = R_{th}$, other wise we compute the power delivered to the load using equation 1.

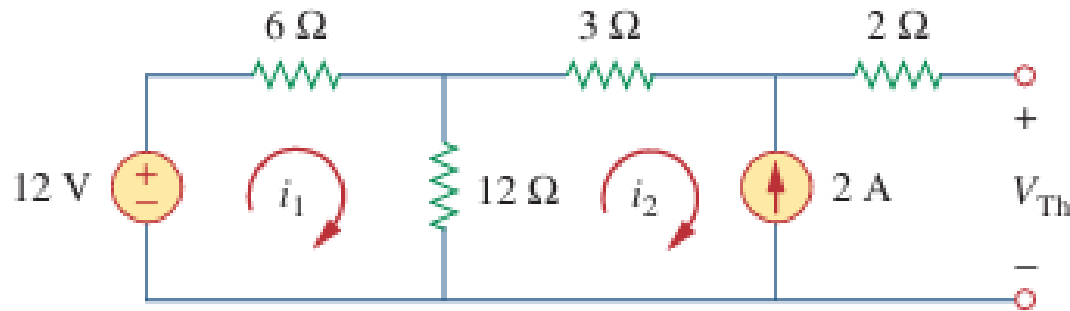
Maximum Power Transfer Theorem

Find the value of R_L for maximum power transfer in the circuit. Find the maximum power.



$$R_{Th} = 2 + 3 + 6 \parallel 12 = 5 + \frac{6 \times 12}{18} = 9 \Omega$$

Maximum Power Transfer Theorem



$$-12 + 18i_1 - 12i_2 = 0, \quad i_2 = -2 \text{ A}$$

$$i_1 = -2/3.$$

$$-12 + 6i_1 + 3i_2 + 2(0) + V_{Th} = 0 \quad \Rightarrow \quad V_{Th} = 22 \text{ V}$$

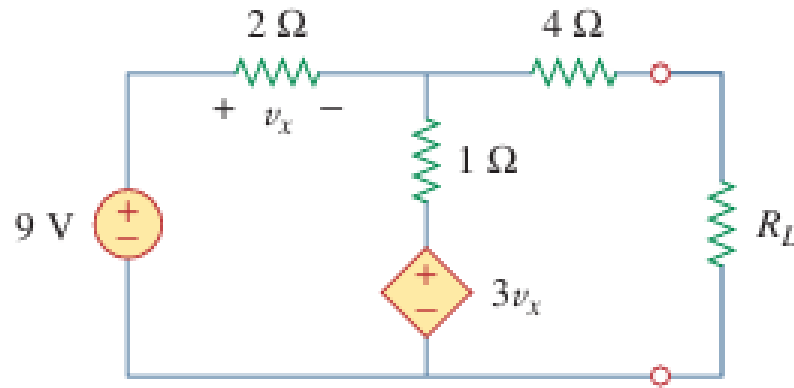
For maximum power transfer $R_L = R_{th}$

$$R_L = R_{Th} = 9 \Omega$$

maximum power is
$$p_{\max} = \frac{V_{Th}^2}{4R_L} = \frac{22^2}{4 \times 9} = 13.44 \text{ W}$$

Maximum Power Transfer Theorem

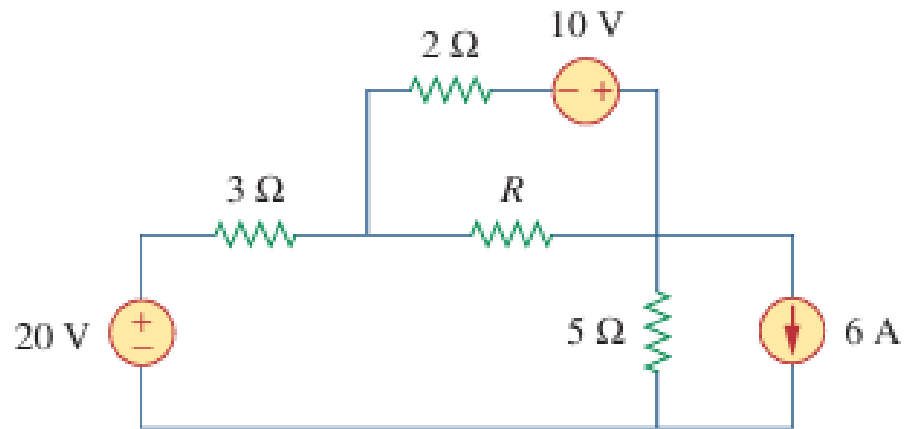
Determine the value of R_L that will draw the maximum power from the rest of the circuit. Calculate the maximum power.



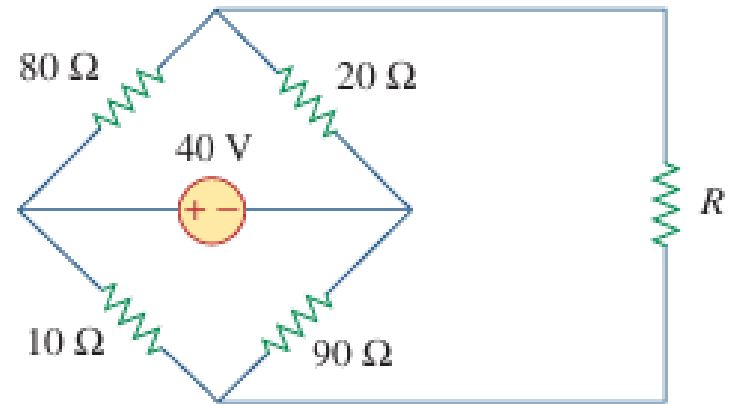
Answer: $R_L = 4.222\ \Omega$, $P_{\max} = 2.901\text{ W}$

Maximum Power Transfer Theorem

Determine the value of R_L that will draw the maximum power from the rest of the circuit (a) & (b). Calculate the maximum power.



(a)



(b)

Next class: Reciprocity Theorem